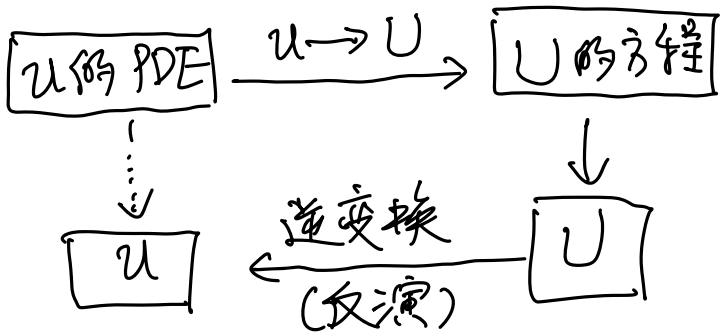
数理方程

第14周
20260529

积分变换法

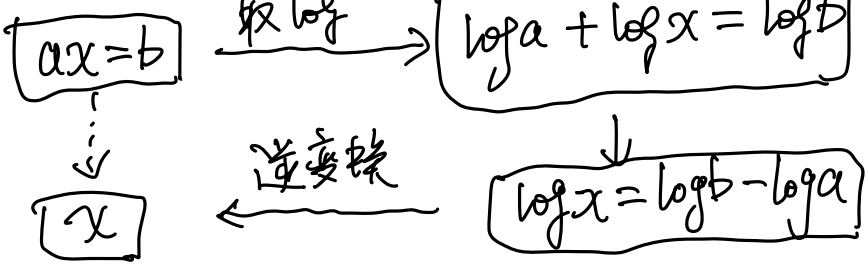
1) Fourier变换（FT）
2) Laplace变换（LT）

微分运算 $\xrightarrow{\text{积分变换}}$ 代数运算



类比：对数

乘除 → 加减



Fourier变换

傅里叶级数 → 傅里叶积分/变换

(1) 级数 $[-L/2, L/2]$

$$f(x) = a_0 + \sum_{n=1}^{\infty}\left(a_n\cos\frac{2\pi n}{L}x + b_n\sin\frac{2\pi n}{L}x\right)$$

正则性

$$a_0 = \frac{1}{L}\int_{-L/2}^{L/2} f(x)\,dx$$

$$a_n = \frac{2}{L}\int_{-L/2}^{L/2} f(x)\cos\frac{2\pi n}{L}x\,dx \qquad b_n = \frac{2}{L}\int_{-L/2}^{L/2} f(x)\sin\frac{2\pi n}{L}x\,dx$$

S-L理论：

$$\begin{cases} y''+\lambda y=0, \quad -\frac{L}{2}<x<\frac{L}{2} \\ y(-\frac{L}{2})=y(\frac{L}{2}), \quad y'(-\frac{L}{2})=y'(\frac{L}{2}) \end{cases} \Rightarrow$$

$\lambda_0=0$, $\lambda_n=(\frac{2\pi n}{L})^2$, $n=1,2,3\cdots$

$X_0=1$, $X_{n1}=\cos\frac{2n\pi x}{L}$, $X_{n2}=\sin\frac{2n\pi x}{L}$

(2) sin, cos → exp $\quad \cos\theta = \frac{e^{i\theta}+e^{-i\theta}}{2}, \quad \sin\theta = \frac{e^{i\theta}-e^{-i\theta}}{2i}$

$$\Rightarrow f(x) = a_0 + \sum_{n=1}^{\infty} a_n \frac{e^{i\frac{2\pi n}{L}x} + e^{-i\frac{2\pi n}{L}x}}{2} + \sum_{n=1}^{\infty} b_n \frac{e^{i\frac{2\pi n}{L}x} - e^{-i\frac{2\pi n}{L}x}}{2i}$$

$$= a_0 + \sum_{n=1}^{\infty} \frac{1}{2}(a_n - ib_n) e^{i\frac{2\pi n}{L}x} + \sum_{n=1}^{\infty} \frac{1}{2}(a_n + ib_n) e^{-i\frac{2\pi n}{L}x}$$

$$= \sum_{n=-\infty}^{\infty} C_n e^{i\frac{2\pi n}{L}x}$$

$$\frac{1}{2}(a_n - ib_n) = \frac{1}{2}\frac{2}{L}\int_{-L/2}^{L/2} f(x)\left(\cos\frac{2\pi n}{L}x - i\sin\frac{2\pi n}{L}x\right)dx = \frac{1}{L}\int_{-L/2}^{L/2} f(x) e^{-i\frac{2\pi n}{L}x}$$

$$C_n = \frac{1}{L}\int_{-L/2}^{L/2} f(x) e^{-i\frac{2\pi n}{L}x} dx \qquad \frac{1}{2}(a_n + ib_n) = \frac{1}{L}\int_{-L/2}^{L/2} f(x) e^{i\frac{2\pi n}{L}x} dx$$

直接采用复本征函数 并由正交性定系数也可得同样结果

$$\rightarrow \int_{-L/2}^{L/2} \left(e^{i\frac{2\pi n}{L}x}\right)^* e^{i\frac{2\pi m}{L}x} dx = \begin{cases} L, & m=n \\ 0, & m\neq n \end{cases}$$

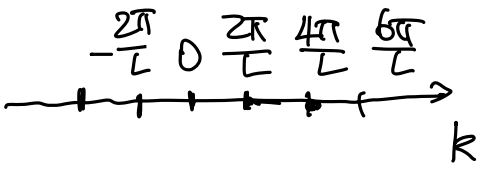
(3) 傅里叶级数 → 傅里叶变换 $\quad L \to \infty$

$$f(x) = \sum_{\substack{k=\frac{2\pi}{L}n \\ n=0,\pm1,\pm2}} \frac{1}{L} F(k) e^{ikx} \qquad C_n \; e^{i\frac{2\pi n}{L}x}$$

$$F(k) = \int_{-L/2}^{L/2} f(x) e^{-ikx} dx$$

$$= \frac{1}{2\pi} \sum_{\substack{k=\frac{2\pi}{L}n \\ n=0,\pm1,\cdots}} \frac{2\pi}{L} F(k) e^{ikx}$$

$\frac{2\pi}{L}$ 小区间长度



$L \to \infty$

$L \to \infty$

$$\Longrightarrow \frac{1}{2\pi}\int_{-\infty}^{+\infty} F(k) e^{ikx} dk$$

$$F(k) = \int_{-\infty}^{+\infty} f(x) e^{-2kx} dx$$

傅里叶变换

$$f(x)=\frac{1}{2\pi}\int_{-\infty}^{+\infty}F(k)e^{ikx}dk$$

傅里叶逆变换（反演）
或 傅里叶积分

$$f(x)\xrightarrow[\text{逆变换}]{\text{傅里叶变换}}F(k)$$

$$\mathcal{F}[f(x)]=F(k)$$

$$\mathcal{F}^{-1}[F(k)]=f(x)$$

为使变换存在且可逆：要求

(1) 任意有限区间上，$f(x)$ 分段光滑

(2) $f(x)$ 在 $(-\infty,\infty)$ 上绝对可积，即 $\int_{-\infty}^{+\infty}|f(x)|dx<\infty$

$\Downarrow$

$f(\pm\infty)=0$

例如 $f(x)=\sin x$

实际应用时，$f(\pm\infty)\neq 0$ 一般也可应用Fourier变换法，技巧是将 $f(x)$ 换成 $f_\epsilon(x)=f(x)e^{-\epsilon|x|}$（$\epsilon>0$），使得 $f_\epsilon(x)$ 绝对可积，得到结果后取 $\epsilon\to 0$.

性质：(1) 线性：

$$C_1f_1(x)+C_2f_2(x)\longleftrightarrow C_1F_1(k)+C_2F_2(k)$$

C_1,C_2：常数

$\Uparrow$

$f_1\longleftrightarrow F_1,\ f_2\longleftrightarrow F_2$

(2) 微分公式：

$$f'(x)\longleftrightarrow ikF(k)$$

$$\mathcal{F}[f'(x)]=\int_{-\infty}^{+\infty}f'(x)e^{-ikx}dx=e^{-ikx}f(x)\Big|_{-\infty}^{+\infty}+ik\int_{-\infty}^{+\infty}f(x)e^{-ikx}dx$$

$e^{-ikx}f(x)\Big|_{-\infty}^{+\infty}=0$

$$=ikF(k)$$

若 $f(\pm\infty)=f'(\pm\infty)=\cdots=f^{(n-1)}(\pm\infty)=0$

$$\Rightarrow f^{(n)}(x)\longleftrightarrow(ik)^nF(k)$$

微分运算 $\longleftrightarrow$ 代数运算

从而减少PDE变量个数，例如PDE $\to$ ODE.

(3) 卷积公式

$f(x)$ 和 $g(x)$ 的卷积：

$$f*g(x)=\int_{-\infty}^{+\infty} f(\xi)\,g(x-\xi)\,d\xi$$

（ξ 对应 $x-\xi$，ξ 对应 ξ）

傅里叶变换卷积公式：← 非常重要，极其常用

$$f*g(x)\longleftrightarrow F(k)\,G(k)$$

证：$\mathcal{F}[f*g(x)]=\int_{-\infty}^{+\infty} dx\, e^{-ikx}\left[\int_{-\infty}^{+\infty} f(\xi)\,g(x-\xi)\,d\xi\right]$

$$=\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} dx\,d\xi\, e^{-ik(x-\xi)}e^{-ik\xi}f(\xi)\,g(x-\xi)$$

$\eta=x-\xi$

$$\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} d\eta\,d\xi\, e^{-ik\eta}e^{-ik\xi}f(\xi)\,g(\eta)=F(k)\,G(k)$$

$$\mathcal{F}^{-1}[F(k)\,G(k)]=f*g(x)$$

卷积的其他例子（不作要求）

例1 骰子A, B. 掷得j点-概率 $\begin{cases} A: a_j \\ B: b_j \end{cases}$ 标准骰子

掷两次点数之和为k的概率

$$c_k = a_1 b_{k-1} + a_2 b_{k-2} + \cdots a_{k-1} b_1$$

$$= \sum_{j=1}^{k-1} a_j b_{k-j}$$ 卷积

记为 $c = a * b$

$$c_2 = a_1 b_1 = \frac{1}{36}$$

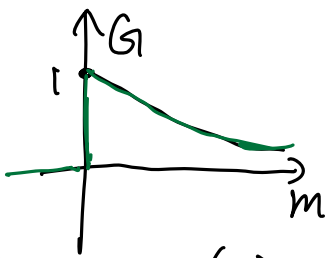
$$c_3 = a_1 b_2 + a_2 b_1 = \frac{2}{36}$$

两个独立随机变量 X, Y: $f_X(x), f_Y(y)$ ← 概率密度

$Z = X + Y$, $f_Z(z) = \int_{-\infty}^{+\infty} f_X(t) f_Y(z-t)\,dt$

$= f_X * f_Y(z)$, 卷积

例2 第k天学得"PDE知识量" p_k. 以 G_m 表示遗忘曲线, 即过了m天后还记得的比例



约定 $G_{m<0} = 0$

第n天"PDE知识总量" $c_n = \sum_{k=-\infty}^{+\infty} p_k G_{n-k}$

$n-k$ 过了 $n-k$ 天

卷积

连续时间

t时刻"知识总量" $C(t) = \int_{-\infty}^{+\infty} G(t-\tau) p(\tau)\,d\tau$

$G(t-\tau)$ ↑ 遗忘曲线

$p(\tau)$ ↑ 学习强度

高维：

三维 $\vec{r}=(x,y,z)$　　$\vec{k}=(k_x,k_y,k_z)$

$$F(\vec{k})=\iiint_{-\infty}^{\infty} f(\vec{r})\,\underbrace{e^{-i\vec{k}\cdot\vec{r}}}_{\uparrow\ e^{-ik_xx-ik_yy-ik_zz}}\,d^3r$$

$$f(\vec{r})\longleftrightarrow F(\vec{k})$$

$$f(\vec{r})=\frac{1}{(2\pi)^3}\iiint_{-\infty}^{+\infty} F(\vec{k})e^{i\vec{k}\cdot\vec{r}}\,\underbrace{d^3k}_{\uparrow dk_xdk_ydk_z}$$

性质：(1) 线性

(2) 微分公式　$\dfrac{\partial f(\vec{r})}{\partial x}\longleftrightarrow ik_xF(\vec{k})$

$$\frac{\partial^2 f(\vec{r})}{\partial x^2}\longleftrightarrow (ik_x)^2F(\vec{k})\ \cdots$$

$$\nabla^2 f(\vec{r})\longleftrightarrow -\vec{k}^2F(\vec{k})$$
$$\uparrow k_x^2+k_y^2+k_z^2$$

(3) 卷积公式

$$f*g(\vec{r})\longleftrightarrow F(\vec{k})G(\vec{k})$$

$$\uparrow \iiint_{-\infty}^{+\infty} f(\vec{\xi})g(\vec{x}-\vec{\xi})\,d^3\xi$$

电势 $u(\vec{r})=\iiint \rho(\vec{\xi})\dfrac{1}{|\vec{r}-\vec{\xi}|}\,d^3\xi$

解PDE

例1

$$\begin{cases}\dfrac{\partial u}{\partial t}-k\dfrac{\partial^2 u}{\partial x^2}=0, & -\infty<x<\infty,\ t>0\\ u|_{t=0}=\phi(x)\end{cases}$$

$$u(x,t)\longleftrightarrow U(k,t)$$
$$=\int_{-\infty}^{+\infty}u(x,t)e^{-ikx}\,dx$$

$$\int_{-\infty}^{+\infty}\left(\frac{\partial u}{\partial t}-k\frac{\partial^2 u}{\partial x^2}\right)e^{-ikx}dx=0$$

$\frac{\partial^2 u}{\partial x^2}e^{-ikx}$ 两次分部积分 $\Rightarrow -k^2U(k,t)$

$\int_{-\infty}^{+\infty}\frac{\partial u}{\partial t}e^{-ikx}dx=\frac{\partial}{\partial t}\left(\int_{-\infty}^{+\infty}ue^{-ikx}dx\right)=\frac{dU(k,t)}{dt}$

$\frac{\partial}{\partial t}$ 可移到对 x 积分号之外

PDE $\Longleftrightarrow$ ODE

$$\frac{dU(k,t)}{dt}+kk^2U(k,t)=0$$

$$U|_{t=0}=\mathcal{F}[\phi(x)]\equiv\Phi(k)$$

$$\Rightarrow U(k,t)=\Phi(k)e^{-kk^2t}$$

$\Phi(k)\leftrightarrow\phi(x)$

$e^{-kk^2t}\leftrightarrow\frac{1}{2\sqrt{\pi kt}}e^{-\frac{x^2}{4kt}}$

$\frac{1}{2\pi}\int_{-\infty}^{+\infty}e^{-kk^2t}e^{ikx}dk=\frac{1}{2\sqrt{\pi kt}}e^{-\frac{x^2}{4kt}}$

$$\int_{-\infty}^{+\infty}e^{-Ak^2-Bk}dk=\int_{-\infty}^{+\infty}e^{-A\left(k+\frac{B}{2A}\right)^2+\frac{B^2}{4A}}dk\xrightarrow{k'=k+\frac{B}{2A}}\int_{-\infty}^{+\infty}e^{-Ak'^2}e^{\frac{B^2}{4A}}dk'=\sqrt{\frac{\pi}{A}}e^{\frac{B^2}{4A}}$$

$B=$ 复数仍成立

$\int_{-\infty}^{+\infty}e^{-Ax^2}dx=\sqrt{\frac{\pi}{A}}$

取 $A=kt$，$B=-ix$

$$\Rightarrow u(x,t)=\mathcal{F}^{-1}[\Phi(k)e^{-kk^2t}]$$

$$=\int_{-\infty}^{+\infty}\phi(\xi)\frac{1}{2\sqrt{\pi kt}}e^{-\frac{(x-\xi)^2}{4kt}}d\xi$$

$G(x,t;\xi,0)$

无界空间热传导问题的格林函数，代表 ξ 处初始温度在 t 时刻对 x 处温度的贡献.

$$\frac{\partial u}{\partial t}=k\frac{\partial^2 u}{\partial x^2} \qquad -\frac{L}{2}<x<\frac{L}{2} \qquad t>0$$

$$u\left(-\frac{L}{2}\right)=u\left(\frac{L}{2}\right),\quad \frac{\partial u}{\partial x}\left(-\frac{L}{2}\right)=\frac{\partial u}{\partial x}\left(\frac{L}{2}\right)$$

$$\Rightarrow X_n''+\lambda_n X_n=0 \quad \Rightarrow \lambda_n=\left(\frac{2\pi n}{L}\right)^2$$

$\frac{2\pi n}{L}\updownarrow k$

$$T_n'(t)+k\left(\frac{2\pi n}{L}\right)^2 T_n(t)=0 \qquad u=\sum_n X_n(x)T_n(t)$$

$$\updownarrow \qquad\qquad \Uparrow$$

$$\frac{dU(k,t)}{dt}+k k^2 U(k,t)=0 \qquad u=\frac{1}{2\pi}\int \Phi(k)e^{-kk^2t}dk$$

分离变量法 $\xrightarrow{L\to\infty}$ 傅里叶变换法

例2:
$$\begin{cases}\dfrac{\partial u}{\partial t}-k\dfrac{\partial^2 u}{\partial x^2}=f(x,t) & -\infty<x<\infty,\ t>0\\ u|_{t=0}=0\end{cases}$$

$u(x,t)\longleftrightarrow U(k,t) \qquad f(x,t)\longleftrightarrow F(k,t)$

$$\Rightarrow \begin{cases}\dfrac{dU(k,t)}{dt}+kk^2U(k,t)=F(k,t)\\ U|_{t=0}\end{cases}$$

$$\Rightarrow U(k,t)=\int_0^t e^{-kk^2(t-\tau)}F(k,\tau)\,d\tau$$

齐次化原理

$$u(x,t) = \mathcal{F}^{-1}\left[\int_0^t e^{-k\xi^2(t-\tau)} F(\xi,\tau)\,d\tau\right]$$

$$= \int_0^t d\tau\ \mathcal{F}^{-1}\left[e^{-k\xi^2(t-\tau)}\ F(\xi,\tau)\right]$$

对t积分与对k积分次序可交换（$\mathcal{F}^{-1}$是对k积分）

$e^{-k\xi^2(t-\tau)}$ ↔ $\frac{1}{2\sqrt{\pi k(t-\tau)}} e^{-\frac{x^2}{4k(t-\tau)}}$

$F(\xi,\tau)$ ↔ $f(x,\tau)$

$$= \frac{1}{2\sqrt{\pi k}} \int_0^t d\tau \int_{-\infty}^{+\infty} d\xi\ f(\xi,\tau) \frac{1}{\sqrt{t-\tau}} e^{-\frac{(x-\xi)^2}{4k(t-\tau)}}$$

卷积公式

三维
$$\begin{cases} \frac{\partial u}{\partial t} - k\nabla^2 u = 0 & -\infty < x, y, z < \infty,\ t > 0 \\ u|_{t=0} = \phi(\vec{r}) & \vec{r} = (x,y,z) \end{cases}$$

$u(\vec{r},t) \longleftrightarrow U(\vec{\xi},t)$, $\phi(\vec{r}) \longleftrightarrow \Phi(\vec{\xi})$

$$\Rightarrow U(\vec{\xi},t) = \Phi(\vec{\xi})\, e^{-k\vec{\xi}^2 t}$$

$\Phi(\vec{\xi})$ ↔ $\phi(\vec{r})$

$e^{-k\vec{\xi}^2 t}$ ↔ $\frac{1}{(4\pi k t)^{3/2}} e^{-\frac{\vec{r}^2}{4kt}}$

$$\Rightarrow u(\vec{r},t) = \frac{1}{(4\pi k t)^{3/2}} \iiint_{-\infty}^{\infty} \phi(\vec{\xi})\, e^{-\frac{|\vec{r}-\vec{\xi}|^2}{4kt}}\, d^3\xi$$

Laplace变换法　　取p足够大，使得积分收敛

$$\mathcal{L}[f(t)] = F(p) = \int_0^\infty e^{-pt} f(t)\,dt$$

拉氏变换存在条件：此积分收敛

$$f(t) = \mathcal{L}^{-1}[F(p)]$$

$$f(t) \longleftrightarrow F(p)$$

只关心 $t \geq 0$：$f(t)$ 理解为 $f(t)H(t)$

单位阶跃函数 $H(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$

由于 e^{-pt} 衰减因子，其收敛性优于傅里叶变换

例：$f(t) = e^{\alpha t}$.　　$\mathcal{L}[e^{\alpha t}] = \int_0^\infty e^{-pt} e^{\alpha t}\,dt = \frac{1}{p-\alpha}$

取 $\mathrm{Re}\,p > \mathrm{Re}\,\alpha$

~~$f(t) = e^{\alpha t^2}$~~ $(\alpha > 0)$ ← 拉氏变换不存在，因为无论多大 p 都无法使积分收敛

性质：1) 线性

2) 微分公式　$f'(t) \longleftrightarrow pF(p) - f(0)$

证：$\mathcal{L}[f'(t)] = \int_0^\infty f'(t) e^{-pt}\,dt = f(t)e^{-pt}\Big|_0^\infty + p\int_0^\infty f(t)e^{-pt}\,dt$

$= -f(0) + pF(p)$

$\mathcal{L}[f'(t)] = p[pF(p) - f(0)] - f'(0)$

$= p^2F(p) - pf(0) - f'(0)$

$$\mathcal{L}[f^{(n)}(t)] = p^n F(p) - p^{n-1}f(0) - p^{n-2}f'(0) - \cdots - pf^{(n-2)}(0) - f^{(n-1)}(0)$$

推论：积分公式 $\mathcal{L}\left[\int_0^t f(\tau)d\tau\right] = \frac{\mathcal{L}[f(t)]}{p} = \frac{F(p)}{p}$

证：$g(t) = \int_0^t f(\tau)d\tau \Rightarrow g'(t) = f(t),\ g(0)=0$

$\Rightarrow \mathcal{L}[g'(t)] = p\mathcal{L}[g(t)] - g(0) = p\mathcal{L}[g(t)]$

$\mathcal{L}[g'(t)] = \mathcal{L}[f(t)] = F(p)$

$\Rightarrow \mathcal{L}[g(t)] = \frac{F(p)}{p}$

(3) 卷积公式 $f * g(t) = \int_0^t f(\tau)\, g(t-\tau)d\tau$

$= \int_{-\infty}^{+\infty} f(\tau)\, g(t-\tau)d\tau$

$f|_{t<0} = g|_{t<0} = 0$

$f(t) \longleftrightarrow F(p)$, $g(t) \longleftrightarrow G(p)$ $\Rightarrow f*g(t) \longleftrightarrow F(p)G(p)$

积分公式可作为卷积公式的推论：

$$\mathcal{L}^{-1}\left[\frac{1}{p}F(p)\right] = \int_0^t 1 \cdot f(\tau)d\tau = \int_0^t f(\tau)d\tau$$

$\mathcal{L}^{-1}\left[\frac{1}{p}\right] = 1$

$\int_0^\infty e^{-pt}\cdot 1\, dt = \frac{1}{p}$

$\mathcal{L}^{-1}[F(p)] = f(t)$

例：$f(t)=1 \longleftrightarrow F(p)=\frac{1}{p}$

$e^{at} \longleftrightarrow \frac{1}{p-a}$

$e^{i\omega t} \longleftrightarrow F(p)=\int_0^{\infty} e^{-(p-i\omega)t}dt=\frac{1}{p-i\omega}$

$e^{-i\omega t} \longleftrightarrow \frac{1}{p+i\omega}$

$\cos\omega t \longleftrightarrow \frac{1}{2}\left(\frac{1}{p-i\omega}+\frac{1}{p+i\omega}\right)=\frac{p}{p^2+\omega^2}$

$\sin\omega t \longleftrightarrow \frac{\omega}{p^2+\omega^2}$

普通反馈以线用到复变函数，略.

ODE：$\begin{cases} y''(t)+\omega^2 y(t)=f(t) & \text{受迫振动} \\ y(0)=A, \quad y'(0)=B \end{cases}$

$y(t)\longleftrightarrow Y(p)$，$f(t)\longleftrightarrow F(p)$

$\Rightarrow p^2Y(p)-p\underbrace{y(0)}_{A}-\underbrace{y'(0)}_{B}+\omega^2Y(p)=F(p)$

$\Rightarrow p^2Y(p)+\omega^2Y(p)=F(p)+Ap+B$

$\Rightarrow Y(p)=\frac{1}{p^2+\omega^2}\left(F(p)+Ap+B\right)$

$\frac{1}{p^2+\omega^2}\longleftrightarrow \frac{1}{\omega}\sin\omega t$，$F(p)\longleftrightarrow f(t)$

$\mathcal{L}^{-1}\left[\frac{p}{p^2+\omega^2}\right]=\cos\omega t$

$\Rightarrow y(t)=\int_0^t \frac{1}{\omega}\sin\omega(t-\tau)f(\tau)\,d\tau+A\cos\omega t+B\frac{\sin\omega t}{\omega}$

PDE $\begin{cases} \frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2} & x>0,\ t>0 \leftarrow \text{细杆振动} \\ \frac{\partial u}{\partial x}\Big|_{x=0} = f(t) \leftarrow \text{端点外力} \\ u|_{t=0} = \frac{\partial u}{\partial t}\Big|_{t=0} = 0 \end{cases}$

已学过的其他方法不易求解此问题

$u(x,t) \longleftrightarrow U(x,p), \quad f(t) \longleftrightarrow F(p)$

$\frac{\partial^2 u}{\partial t^2} \longleftrightarrow p^2 U - p\,u|_{t=0} - \frac{\partial u}{\partial t}\Big|_{t=0} = p^2 U$

$\Rightarrow \begin{cases} p^2 U(x,p) = a^2 \frac{d^2}{dx^2} U(x,p) \\ \frac{dU}{dx}\Big|_{x=0} = F(p), \quad U|_{x=+\infty} = 0 \end{cases}$

$\Uparrow$ $u|_{x\to\infty} = 0$, 因端点影响的传播速度有限

$\Rightarrow U(x,p) = \cancel{C e^{\frac{p}{a}x}} + D e^{-\frac{p}{a}x}$ $\Leftarrow D = -\frac{a}{p}F(p)$

$= \left(-\frac{a}{p}\right) e^{-\frac{p}{a}x} F(p)$

A B C

(AB)C, (AC)B, (BC)A 三种组合方式

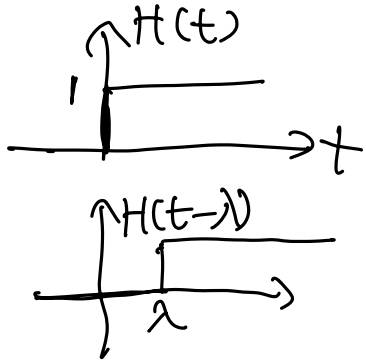
$\Rightarrow$ 三种反演方式

① (AB)C方式，反演AB和C，再卷积

↓

反演A和B，再卷积

$$H(t)=\begin{cases}1, & t\geq 0\\ 0 & t<0\end{cases}$$



查表 $\mathcal{L}^{-1}\left[\frac{1}{p}e^{-\lambda p}\right]=H(t-\lambda)$

$$\mathcal{L}[H(t-\lambda)]=\int_0^\infty e^{-pt}H(t-\lambda)\,dt$$

$$=\int_\lambda^\infty e^{-pt}\,dt=\frac{1}{p}e^{-\lambda p}$$

$\lambda=\frac{x}{a}$

$$\Rightarrow \mathcal{L}^{-1}\left[-\frac{a}{p}e^{-\frac{x}{a}p}\right]=-aH\left(t-\frac{x}{a}\right)$$

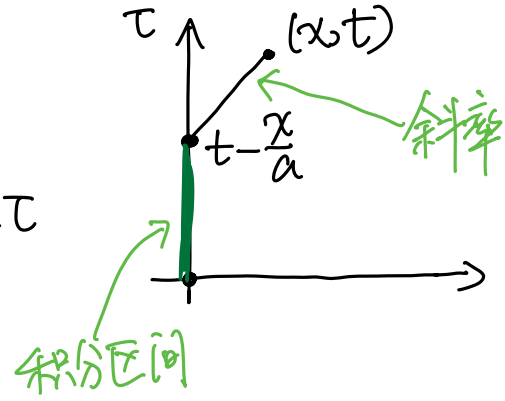
$$\Rightarrow u(x,t)=\mathcal{L}^{-1}\left[-\frac{a}{p}e^{-\frac{x}{a}p}\cdot F(p)\right]$$

$$=\int_0^t -aH\left(t-\tau-\frac{x}{a}\right)f(\tau)\,d\tau$$

$$H\left(t-\tau-\frac{x}{a}\right)=\begin{cases}1 & \tau\leq t-\frac{x}{a}\\ 0 & \tau>t-\frac{x}{a}\end{cases}$$

$$=\begin{cases}0, & t<\frac{x}{a}\\ -a\int_0^{t-\frac{x}{a}}f(\tau)\,d\tau, & t\geq\frac{x}{a}\end{cases}$$

$$=-aH\left(t-\frac{x}{a}\right)\int_0^{t-\frac{x}{a}}f(\tau)\,d\tau$$



$\int_0^\infty e^{-pt} g(t-\lambda)H(t-\lambda)dt = \int_\lambda^\infty e^{-pt} g(t-\lambda)dt$

$= \int_\lambda^\infty e^{-p(t-\lambda)} e^{-p\lambda} g(t-\lambda)dt$

$\xrightarrow{t'=t-\lambda} e^{-p\lambda}\int_0^\infty e^{-pt'} g(t')dt'$

$= e^{-\lambda p}G(p)$

② (AC)B方式，反演AC和B

$$\mathcal{L}^{-1}\left[-\frac{a}{p}F(p)\right] = -a\int_0^t f(\tau)d\tau$$

积分公式

$\leftarrow \lambda > 0$

查表有 $\mathcal{L}^{-1}[e^{-\lambda p}G(p)] = g(t-\lambda)H(t-\lambda)$ ← 称为"延迟定理"

取 $\lambda = \frac{x}{a}$，$G(p) = -\frac{a}{p}F(p) \Longleftrightarrow g(t) = -a\int_0^t f(\tau)d\tau$

$$\Rightarrow \mathcal{L}^{-1}\left[-\frac{a}{p}F(p)e^{-\frac{x}{a}p}\right] = H\left(t-\frac{x}{a}\right)g\left(t-\frac{x}{a}\right)$$

$$= H\left(t-\frac{x}{a}\right)\left(-a\int_0^{t-\frac{x}{a}} f(\tau)d\tau\right)$$

$$= -aH\left(t-\frac{x}{a}\right)\int_0^{t-\frac{x}{a}} f(\tau)d\tau$$

③ (BC)A方式.

查表有 $\mathcal{L}^{-1}[F(p)e^{-\frac{x}{a}p}] = H\left(t-\frac{x}{a}\right)f\left(t-\frac{x}{a}\right)$

↑ 延迟定理

$$\Rightarrow \mathcal{L}^{-1}\left[-\frac{a}{p}F(p)e^{-\frac{x}{a}p}\right] = -a\int_0^t H\left(\tau-\frac{x}{a}\right)f\left(\tau-\frac{x}{a}\right)d\tau$$

↑ 积分公式

$\xrightarrow{\tau'=\tau-\frac{x}{a}}$ $\begin{cases} -a\int_0^{t-\frac{x}{a}} f(\tau')d\tau', & t \geq \frac{x}{a} \\ 0, & t < \frac{x}{a} \end{cases}$ $= -aH\left(t-\frac{x}{a}\right)\int_0^{t-\frac{x}{a}} f(\tau)d\tau$

PDE求解问题里，Laplace变换一般是对时间t做.

常用来求解初值问题，即已知$t=0$时刻函数值，求$t>0$时的函数.

除了Fourier、Laplace，还有其他多种积分变换.

例：正弦变换 $F(k)=\int_0^\infty f(x)\sin kx\,dx$

$f(x)=\frac{2}{\pi}\int_0^\infty F(k)\sin kx\,dk$

一般形式：$F(k)=\int_a^b \underbrace{K(k,x)}_{\text{核}} f(x)\,dx$

$$K(k,x)=\begin{cases} e^{-kx}, & 0\le x<\infty & \text{Laplace} \\ e^{-ikx} & -\infty<x<\infty & \text{Fourier} \\ \sin kx & 0\le x<\infty & \text{正弦} \\ \cos kx & 0\le x<\infty & \text{余弦} \\ xJ_n(kx) & 0\le x<\infty & \text{Hankel} \\ x^{k-1} & 0\le x<\infty & \text{Mellin} \end{cases}$$

少数情况下，直接利用逆变换表达式比卷积定理快.

傅里叶变换

$$\phi(x) \longleftrightarrow \Phi(k)$$

$$? \longleftrightarrow \Phi(k)e^{ikat}$$

可直接根据逆变换公式：

$$\frac{1}{2\pi}\int_{-\infty}^{+\infty} \Phi(k)e^{ikat}e^{ikx}dk = \frac{1}{2\pi}\int_{-\infty}^{+\infty} \Phi(\phi)\, e^{ik(x+at)}dk$$

$$= \phi(x+at)$$

熟悉 Dirac δ 函数的同学，也可用卷积定理计算：

由 $\delta(x+x_0) \longleftrightarrow e^{ikx_0}$，取 $x_0 = at$

$$\Rightarrow \mathcal{F}^{-1}\left[\Phi(k)e^{ikat}\right] = \int_{-\infty}^{+\infty} \phi(\xi)\,\delta(x-\xi+at)\,d\xi$$

$$= \phi(x+at)$$

练习：
$$\begin{cases} \dfrac{\partial^2 u}{\partial t^2} - a^2 \dfrac{\partial^2 u}{\partial x^2} = 0, \quad -\infty < x < \infty, \ t > 0 \\ u|_{t=0} = \phi(x), \quad \left.\dfrac{\partial u}{\partial t}\right|_{t=0} = \psi(x) \end{cases}$$

解：$u(x,t) \longleftrightarrow U(k,t)$，$\phi(x) \longleftrightarrow \Phi(k)$，$\psi(x) \longleftrightarrow \Psi(k)$

$$\Rightarrow \begin{cases} \dfrac{d^2 U(k,t)}{dt^2} + a^2 k^2 U(k,t) = 0 \\ U|_{t=0} = \Phi(k), \quad \left.\dfrac{dU}{dt}\right|_{t=0} = \Psi(k) \end{cases}$$

$$\Rightarrow U(k,t) = \Phi(k)\cos kat + \Psi(k)\frac{\sin kat}{ka}$$

$$\Rightarrow u(x,t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} dk\, U(k,t) e^{ikx} dk$$

$$= \frac{1}{2\pi}\int_{-\infty}^{+\infty} dk \left[\Phi(k)\cos kat + \Psi(k)\frac{\sin kat}{ka}\right] e^{ikx}$$

第一项 $= \frac{1}{2\pi}\int_{-\infty}^{+\infty} \Phi(k)\frac{e^{ikat}+e^{-ikat}}{2}e^{ikx}\,dk$

$= \frac{1}{2\pi}\int_{-\infty}^{+\infty} \Phi(k)\frac{e^{ik(x+at)}+e^{ik(x-at)}}{2}\,dk$

$\phi(x+at)$，因为 $\frac{1}{2\pi}\int_{-\infty}^{+\infty}\Phi(k)e^{ikx}dk=\phi(x)$，其中 x 换成 $x+at$ 即可

$= \frac{1}{2}[\phi(x+at)+\phi(x-at)]$

第二项 $= \frac{1}{2\pi}\int_{-\infty}^{+\infty}\Psi(k)\frac{\sin kat}{ka}e^{ikx}dk$

$= \frac{1}{2\pi}\int_{-\infty}^{+\infty}\Psi(k)\int_0^t \cos ka\tau\,d\tau\; e^{ikx}\,dk$

$= \frac{1}{2\pi}\int_0^t d\tau\int_{-\infty}^{+\infty}dk\;\Psi(k)\frac{e^{ik(x+a\tau)}+e^{ik(x-a\tau)}}{2}$

$= \frac{1}{2}\int_0^t d\tau\,[\psi(x+a\tau)+\psi(x-a\tau)] = \frac{1}{2}\left[\frac{1}{a}\int_x^{x+at}\psi(\xi)d\xi - \frac{1}{a}\int_x^{x-at}\psi(\xi)d\xi\right]$

$\xi = x+a\tau$，$\xi = x-a\tau$

$= \frac{1}{2a}\int_{x-at}^{x+at}\psi(\xi)d\xi$

$\Rightarrow u(x,t) = \frac{1}{2}[\phi(x+at)+\phi(x-at)] + \frac{1}{2a}\int_{x-at}^{x+at}\psi(\xi)d\xi$

	Fourier	Laplace
变量定义域	$(-\infty, \infty)$	$[0, \infty)$
IC	手动加入	变换为程自动包含IC
BC	全空间	变换后BC原样保留
逆变换难度	相对简便	常需复变函数计算或查表
典型应用场景	无界空间	时间演化初值问题